

SYSEN 5151 Project Progress Milestone Review Criteria

Milestone Schedule

Milestone	Timing	Primary Review Purpose	Expected Project Maturity
Milestone Check 1	October 7, 2026	Confirm that the team has a defensible Sol definition, stakeholder needs and requirements baseline, and initial product implementation.	MBSE Model Build: Sol definition and stakeholder needs and requirements baseline established. Product Build: walking skeleton demonstrated.

Milestone Check 1 — System Definition and Requirements Baseline

October 7, 2026

This checkpoint occurs after the course covers business/mission analysis, operational context and behavior, stakeholder needs, requirements definition, requirements quality/traceability, and as the class begins top-level functional architecture. The purpose is to ensure that teams have defined what they are building, why it is needed, and what it must do before the product grows beyond a controlled walking skeleton.

A. MBSE Model / Systems Entrance Criteria

- A clearly stated project mission/problem and System of Interest (SoI) are established.
- The operational concept (OpsCon) is modeled, and the system boundary is explicit.
- All significant external actors/systems that directly interact with the SoI are identified.
- Context, hierarchy, and spider diagrams are internally consistent and show the relationship between form and function.
- The primary operational use case is modeled, with appropriate use-case and behavior views (e.g., use case, action/activity, and sequence diagrams as required by the lab sequence).
- Stakeholder needs and requirements are written using the course requirement-quality rules.
- Requirements are traceable back to stakeholder needs & each accepted requirement has a measurable or testable acceptance criterion.
- System-level functional requirements beginning to emerge are identified.

B. Product / API Development Entrance Criteria

- The GitHub repository structure is established and consistent with the system context rather than being an unrelated generated starter project.
- README/context documentation reflects the approved OpsCon and external-system inventory.
- A walking skeleton demonstrates one primary end-to-end system path; stubs are acceptable at this stage.
- A specification file (e.g., SPEC.md) is being developed from the approved needs/requirements rather than from a loose product idea.
- GenAI-assisted work is recorded in the project prompt/provenance log as required by the course.
- AI code generation is constrained by requirement/specification text where applicable; students should not use a broad prompt to generate the complete project.

C. Milestone 1 Pass Exit Criteria

- The team can clearly answer: What is the SoI, who needs it, what problem does it address, what lies inside/outside the boundary, and what must it do?
- A reviewer can trace at least one primary stakeholder need to a stakeholder requirement and then to the modeled use case/function it motivates.
- The walking skeleton is demonstrably linked to the modeled primary use case.
- There are no major unexplained software capabilities that lack a corresponding modeled need/requirement/use case.
- The team is ready to proceed into detailed functional/physical architecture without redefining the fundamental project scope.

Milestone 1 Review Checklist

Review Area	Required Evidence / Completion Criterion	Status	Reviewer Notes
Mission / Sol / Boundary	Mission/problem, Sol, system boundary, and external systems are clear and consistent.	☐ Met ☐ Partial ☐ Not Met	
OpsCon & Behavior	OpsCon plus primary use case/behavior views are complete enough to explain system operation.	☐ Met ☐ Partial ☐ Not Met	
Stakeholder Needs	Stakeholders and integrated needs are documented and defensible.	☐ Met ☐ Partial ☐ Not Met	
Requirements Baseline	Well-formed stakeholder requirements exist with need-to-requirement traceability and acceptance criteria.	☐ Met ☐ Partial ☐ Not Met	
Walking Skeleton	One end-to-end path runs, even if some participants are stubbed.	☐ Met ☐ Partial ☐ Not Met	
Model-to-Product Linkage	Repository/specification/walking skeleton can be linked back to the MBSE model.	☐ Met ☐ Partial ☐ Not Met	